Fiscal Year:	FY 2022	**Task Last Updated:**	FY 08/17/2021
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	LOOP-based Constrained Vapor Bubbles		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-E: Use of the NASA Physical Sciences Informatics System – Appendix E
Start Date:	11/01/2019	**End Date:**	10/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2022 per PI; original end date was 10/31/2021 (Ed., 8/17/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC20K0120		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Loop heat pipes are quite common heat transfer devices in satellites and space systems but often fail and the causes are not always readily apparent. We had built a prototype loop heat pipe as part of the original Constrained Vapor Bubble (CVB) program but only used it to look at refrigerants and especially to help understand meniscus stability. In the proposal, we would like to resurrect that system, based on a wickless, square cross-section CVB and use it to collect ground-based data that we can compare with the linear heat pipe CVB data obtained during CVB1 and CVB2 missions. In particular, we are concerned with mapping the vapor liquid interface, driving the system to the point where Marangoni flows become important, and understanding the mechanisms of failure or instability in wickless loop systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work will enhance the development of heat pipes which are not only used for space applications but also for cooling high heat generation devices on Earth. We are also augmenting and enhancing our interferometry techniques which will greatly expand the types of fluids and surfaces that we can investigate . This will have great utility in looking at the fundamental mechanisms governing phase change heat and mass transfer. Since phase change is a key process in distillation, power generation, refrigeration, and electronics cooling, the more we can learn and uncover the more efficient devices we will be able to manufacture.
Task Progress:	During the course of this work we have been vigorously working on designing the loop system. In parallel we have been designing a linear heat pipe for a CASIS (Center for the Advancement of Science in Space) project. COVID interrupted some of our research effort, so we were unable to send the part out for fabrication for some time. When we were ready, we received the linear heat pipe and promptly broke the glass-to-metal seal when we tried to install the pressure transducer. So, we placed the loop order on hold until we were confident we could mate the transducer to the heat pipe without damaging the system. The problem has been corrected and so the loop fabrication will be sent out August 2021 and we should begin testing it by October. In the meantime, we have been doing several theoretical studies and reinvestigating some of the older data from CVB-1. In the first study, we unified results from our heater power variation and condenser temperature variation studies to show that the Nusselt number characterizing the evaporation section of the CVB was directly correlated with the Marangoni number. The relationship was to the inverse 1/3 power so Marangoni flows in this system did not aid heat transfer. As we were working under Covid restrictions, we also had a chance to look into our optical techniques a bit more. We were interested in the effects of surface modification on the Hamaker constant. Many years ago, we looked into physical modifications via deposition or etching to produce nanostructures and showed the effect of such formations on the Hamaker constant. This time, with the advent of atomic layer deposition techniques, we became interested in planar modifications using high index materials like Hafnium Dioxide and Silicon Nitride. When films of these materials or SiO2 were deposited on the Si substrates, it became clear that the old techniques we had been using to analyze our interferometry data and especially the regions around the thin, adsorbed film were inadequate. So, we have been developing a technique whereby we analyze the images captured using reflectivity software (Scout) that, with a little calibration for camera settings, can handle any scenario, provide data on the adsorbed film thickness and the transition region between the minimum thickness we can measure using interferometry and the adsorbed film, and provide us with a better understanding of the width of the contact line. We are deploying that technique now for both the linear and loop heat pipes as well as a capillary feeding apparatus we developed to easily experiment with our modified surfaces. Lastly, we are continuing to see if we can simulate the entirety of the fringe image patterns we obtain using finite element techniques; basically solving Maxwell's equations to predict the intensity profiles of the images we see in the microscope. In a second study, we returned to some of the strangest CVB-1 behavior, heterogeneous nucleation in the 20 mm module. A single nucleation event was studied and published in the International Journal of Heat and Mass Transfer in 2021 (see Bibliography section). In this current paper, we looked at over 20 such events, culled from over 100 that we observed, but spanning the complete range of system pressures and superheats that led to, or were the result of, nucleation. The main result was that we could correlate all the heat transfer data using a single equation relating the Nusselt number to the Marangoni number and the reduced pressure of the nucleation event.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	Yu J, Nguyen TTT, Pawar A, Wayner PC Jr, Plawsky JL, Chao DF, Sicker RJ. "The effect of condenser temperature on the performance of the evaporator in a wickless heat pipe performance." International Journal of Heat and Mass Transfer. 2021 Sep;176:121484. Available online 1 June 2021. https://doi.org/10.1016/j.ijheatmasstransfer.2021.121484 , Sep-2021
Papers from Meeting Proceedings	Pawar A, Plawsky JL. "Computational Study of Oscillating 3-phase Contact Line on Flat Surfaces." ASME 2021 Summer Heat Transfer Conference, Virtual, June 16-18, 2021. ASME 2021 Summer Heat Transfer Conference, Virtual, June 16-18, 2021. Technical presentation SHTC2021-63268. , Jun-2021
Papers from Meeting Proceedings	Yu J, Pawar A, Wayner P, Plawsky J, Chao D, Sicker R. "The Effect of Condenser Temperature on the Performance of the Evaporator in a Wickless Heat Pipe Performance." ASME 2021 Summer Heat Transfer Conference, Virtual, June 16-18, 2021. ASME 2021 Summer Heat Transfer Conference, Virtual, June 16-18, 2021. Technical presentation. SHTC2021-63137. , Jun-2021